

Rowing Metrics

User Guide — Screens & Options

Cross-platform rowing session tracker for Android and iOS.

This document describes every screen, metric, and configuration option in the app.

1. Overview

Rowing Metrics tracks rowing sessions using the phone accelerometer for stroke rate (SPM) and GPS for speed and distance. The same interface runs on Android and iOS (shared Kotlin Multiplatform UI).

Key capabilities:

- Live stroke rate, speed, average speed, distance, and elapsed time
- Configurable stroke detection sensitivity and GPS speed smoothing
- Automatic saving of completed sessions to local history
- Export activity history to CSV (Excel-compatible) on Android
- UI languages: English, Catalan, Spanish, and French

Navigation

The app has four screens in a horizontal pager. Swipe left or right to move between them:

#	Screen	Purpose
1	Main metrics	Live session display — Start / Stop
2	Activities	History of completed sessions; export / delete
3	Configuration	Stroke sensitivity, speed smoothing, language
4	About	App name, version, and contact

2. Main Metrics Screen (Screen 1)

Primary on-water display. Shows live metrics while a session is running, and final values after Stop. Layout adapts to portrait and landscape.

Metrics displayed

Field	Unit / format	Description
Stroke rate	SPM (integer)	Strokes per minute from accelerometer. Largest value on screen.
Speed	km/h (1 decimal)	Current GPS speed, smoothed by the Speed smoothing setting.
Average speed	km/h (1 decimal)	Total distance ÷ elapsed time for the session.
Distance	meters (integer)	Cumulative GPS distance since Start (segments "e 4 m).
Time	MM:SS or HH:MM:SS	Elapsed session time. Shows hours from 1 hour onward.
Clock	HH:MM	Device local time (updates every second), beside Time.

Start / Stop button

Green Start begins a session. While running it turns red and shows Stop.

- Start: accelerometer stroke detection, GPS tracking, and session timer
- Stop: ends the session and saves a summary to Activities
- On first Start, Android requests location permission if not already granted

Stroke detection warning

While running, if no stroke has been detected recently, the stroke rate band blinks between light red and black. Adjust phone placement or increase detection sensitivity in Configuration.

Portrait layout

Top to bottom: Stroke rate, Speed, Average speed, Distance, Time + Clock, Start/Stop.

Landscape layout

Left: Stroke rate and Start/Stop. Right: Speed + Average speed, Distance, Time + Clock.

Android-specific behavior

- Screen stays on while the app is in the foreground
- Can appear over the lock screen and turn the screen on while rowing
- Requires accelerometer/gyroscope for stroke detection

3. Activities Screen (Screen 2)

Title: “Activities”. Scrollable table of completed sessions saved when you press Stop.

Table columns

Column	Content
Date	Date the session ended
Hour	Start time (end time " duration)
Time	Session duration
Stroke rate	Average SPM for the session
Avg. speed	Average speed in km/h
Distance	Total distance in meters
Delete	Remove that single session

Empty state

“No activities yet. Press Start on the first screen, then Stop to record one.”

Export activities (Android)

- Download icon appears when at least one activity exists
- Confirm “Export to Excel”, then choose where to save the CSV
- Columns: Date, Hour, Time, Stroke rate, Avg. speed (km/h), Distance

Delete all

“Delete all” asks for confirmation and permanently removes every session.

Delete one session

Tap Delete on a row to remove that session immediately.

4. Configuration Screen (Screen 3)

Title: “Rowing Metrics Configuration”. Changes are saved immediately. Stroke sensitivity applied to the next session (or after Stop if changed mid-session).

Stroke profile — Detection sensitivity

5-step horizontal slider. Left = strictest (Low); right = most permissive (High). Only Low, Medium, and High show text labels; intermediate steps are tick marks only.

Position	Label	Behavior
1 (left)	Low	Strictest — fewer false strokes; soft strokes may be missed
2	(tick)	Between Low and Medium
3 (center)	Medium	Default — balanced
4	(tick)	Between Medium and High
5 (right)	High	Most permissive — easier detection; more false positives

Move toward High if strokes are missed; toward Low in rough water or when handling the phone causes false counts.

Speed smoothing

Slider 1–8 (default 4). How many recent filtered GPS speed samples are averaged for live Speed. Does not change Average speed (always distance ÷ time).

Language

Dropdown: English (default), Catalan, Spanish, French. Applies to all UI labels and CSV headers.

5. About Screen (Screen 4)

Shows app title "Rowing Metrics", version number, and contact gardenyes@gmail.com.

6. How Metrics Are Calculated

Meaning of the numbers behind the UI.

Stroke rate (live SPM)

- Accelerometer samples detect stroke peaks with adaptive thresholds
- SPM updates about every 2 seconds from recent stroke intervals
- Needs several strokes before a stable SPM is shown
- Returns toward 0 if no stroke is detected for a few seconds
- Session average SPM = mean of SPM samples while running (not strokes ÷ time)

Speed (current)

- GPS speed is filtered, then averaged over the last N samples (N = Speed smoothing)
- Shown in km/h; poor GPS fixes are rejected

Average speed

Total session distance ÷ elapsed time (not an average of instantaneous speeds).

Distance

- From GPS position changes
- Only segments of at least about 4 meters are counted

Orientation changes

If phone orientation changes during a session, stroke detection pauses briefly (~1.2 s) then resumes.

7. Requirements & Platform Notes

Permissions (Android)

- Location (Fine or Coarse): for speed and distance
- Motion sensors: for stroke detection

Platform differences

Feature	Android	iOS
Stroke detection	Full support	Pending Core Motion bindings
GPS speed / distance	Full support	Pending Core Location bindings
Activities history	Full support	Full support
Configuration / language	Full support	Full support
CSV export	Full support	Not yet implemented
Lock screen display	Yes	Platform default

Hardware

- GPS for speed and distance
- Accelerometer / gyroscope for stroke rate

8. Quick Reference

Screen swipe order

Main metrics !' Activities !' Configuration !' About

Default settings

Setting

Detection sensitivity

Speed smoothing

Language

Default

Medium (center)

4

English

Typical workflow

1. Open the app (grant location when prompted).
2. Optionally open Configuration to adjust sensitivity or language.
3. On the main screen, press Start when you begin rowing.
4. Monitor stroke rate, speed, and distance live.
5. Press Stop when finished — the session is saved to Activities.
6. Review, export, or delete past sessions on Activities.